

Experiment 3

Code:

```
# ID3 Decision Tree Algorithm

from sklearn.tree import DecisionTreeClassifier, export_text

# Training data

X = [
    [0,0,0],
    [0,0,1],
    [1,0,0],
    [2,1,0],
    [2,2,0],
    [2,2,1],
    [1,2,1],
    [0,1,0],
    [0,2,0],
    [2,1,0],
    [0,1,1],
    [1,1,0],
    [1,0,1],
    [2,1,1]
]

# Target (Play Tennis: Yes=1, No=0)

y = [0,0,1,1,1,0,1,0,1,1,1,1,1,0]

# Train ID3 Decision Tree

clf = DecisionTreeClassifier(criterion="entropy")

clf.fit(X, y)

# Print Decision Tree

print(export_text(clf, feature_names=["Outlook","Temperature","Humidity"]))

# Classify a new sample
```

```
new_sample = [[0,1,0]]
prediction = clf.predict(new_sample)
print("Prediction:", "Yes" if prediction[0] == 1 else "No")
```

Output:

```
▶ print("Prediction:", "Yes" if prediction[0] == 1 else "No")
```

```
... |--- Outlook <= 0.50
    | |--- Temperature <= 0.50
    | | |--- class: 0
    | |--- Temperature > 0.50
    | | |--- Humidity <= 0.50
    | | | |--- Temperature <= 1.50
    | | | |--- class: 0
    | | | |--- Temperature > 1.50
    | | | |--- class: 1
    | | |--- Humidity > 0.50
    | | |--- class: 1
    |--- Outlook > 0.50
    | |--- Humidity <= 0.50
    | | |--- class: 1
    | |--- Humidity > 0.50
    | | |--- Outlook <= 1.50
    | | | |--- class: 1
    | | |--- Outlook > 1.50
    | | |--- class: 0
```

Prediction: No